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1. **Find out and explain how a hydrofoil works so it can lift a vessel out of the water, and what's the purpose**

   Hydrofoil works as a **foil** operating in water. At low speeds, the hull (body of ship) rests in the water, but as the **speed** encreases the hydrofoil **lifts** the vessel. As the lift matches the weight of the boat, the whole hull comes out from the water.

   Looking at **Bernoulli's Equation**

$$Po = P_1 + \frac{1}{2}\rho v_1^2 + \rho g y_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho g y_2 \quad \dots (1)$$

   Where Po is the stagnation pressure (Po), P1 and P2 is pressure (Pa), rho is the medium density (kg/m^3), y is the height, and v is velocity (m/s). Figure one shows the line flow of the water through a hydrofoil.

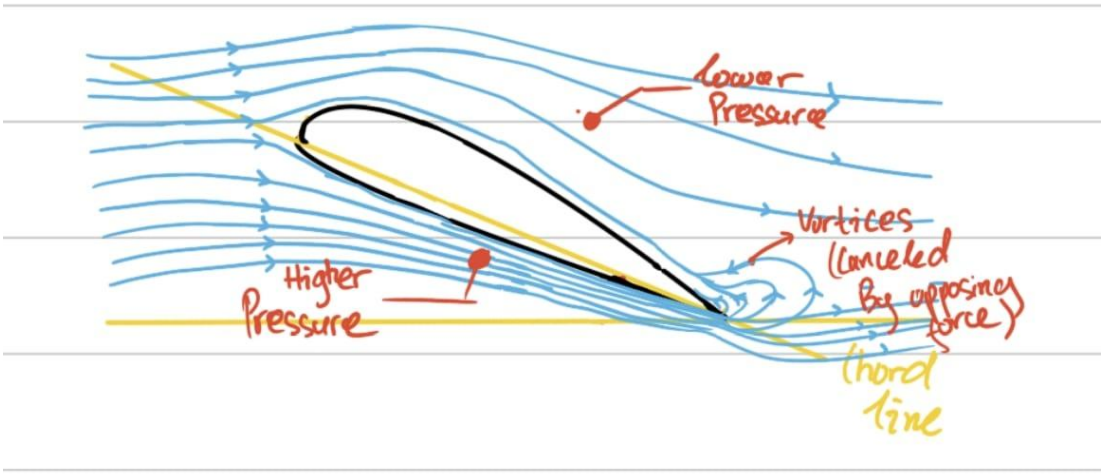


Figure 1. Bernoulli's Principle Visualization (Src: Personal Documentation)

   The height difference between the two point is approximate to the thickness of the foil which makes it influence negligible. Rewriting Eq. (1) we will get

$$P_1 + \frac{1}{2}\rho v_1^2 = P_2 + \frac{1}{2}\rho v_2^2 \quad \dots (2)$$

   The equation descrbing the flow of water through point 1 and 2 in Fig. 1. The Bernoulli's Equation dictates that **the increase of a fluid's velocity occours simultaneously with the the decrease in its static pressure**. The water under the foil has less velocity because of **viscous effecs** (friction with the foil surface) which lead to the formation of **vortices** (rotating movement caused by low pressure once the liquid under reaches the trailing edge and exposed to the low pressure above). Rotation

creates angular momentum, and to conserve it, we need the equal and opposite amount of momentum exchange to the vortex. The vector summation of the veolcities results in a **higher speed on top**. Applying what we've know to Eq. (2), we see that to keep both sides equal as its stagnation pressure, **P1 is higher than P2**. Since force is proportional to pressure, it means that **theres a net force going up from the hydrofoil which we call lift**.

We could also prove the lift's existence by looking at **Euler's Equation for Flow Normal (Perpendicular to the foil surface) to a Streamline**

$$\frac{d}{dn}(P + \rho g y) = \frac{\rho v^2}{R} \dots (3)$$

Which is equal to the the fluid density times velocity squared devided by R. As the water on top going through the curved surface, it creates a curved movement that is traced by interpreting it as a circular motion with radius R as we could visualize through Fig 2. Keep in mind that the normal (n) here has a direction facing away from the center of the circular motion and is relatively facing away from the foil surface. **The more it curves through the top foil, the smaller the R, hence the higher the pressure change along the normal surface and therefore the foil is lifted upwards**.

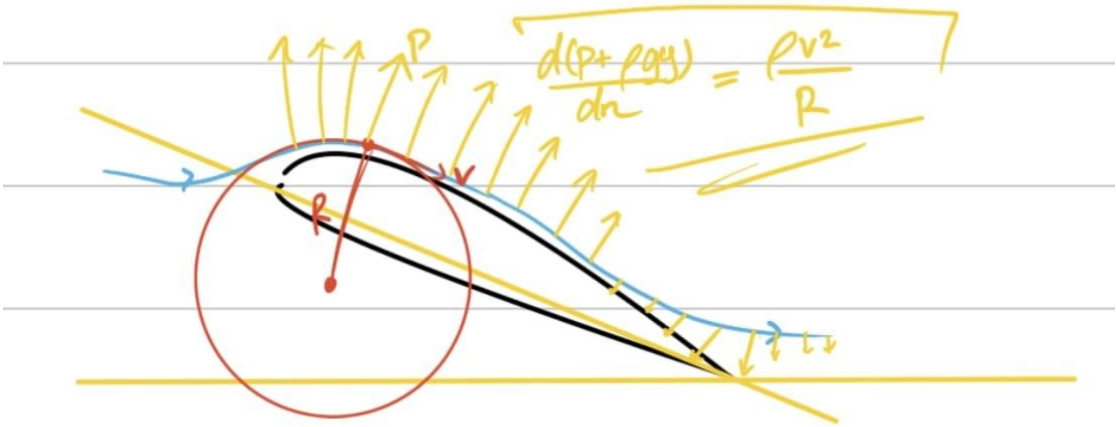


Figure 2. Euler's Equation Visualization (Src: Personal Documentation)

In conventional boats, the more it speeds up, water molecules are crashing to the big surface of the hull, creating an opposite force called **drag**. By lifting up the hull, we've eliminated the drag produced by the hull, hence making it more **efficient and fast** cruising. It contributes also to the **power** needed for the boat. Hydrofoil boats are also does not create divergent and transverse waves, which is the waves that usually ripples from a moving conventional boat. Having the hull lifted also smooths out movement by avoiding waves from the water. Until today, hydrofoils is used for high-speed and

efficient Until today, hydrofoils is used for high-speed and efficient water transportation, but in recent years, the technology was also implemeted on a surfboad, creating **hydrofoiling**.

2. **Active Hydrofoil allows us to change the foil's angle of attack. What's the benefit? What is the effect of the foil angle on the vessel's height and orientation stability?**
   Active hydrofoil uses **control system** to control a movable **foil flaps** to stabilize the vessel's orientation: **heave (up-down), roll (front-back axis of angle), and pitch (left-right axis of angle)** so that **it could transition from the hullborn (submerged hull) to foilborne (lifted hull) and ready for cruising**. The maximum lift:drag ratio is 20 to 25:1, by designing a control system with a setpoint of that ratio, we can control the foil flaps to adjust its angle of attack. A **Proportional, Integral, and Differential (PID) loop** is one of the systems that could be used here.

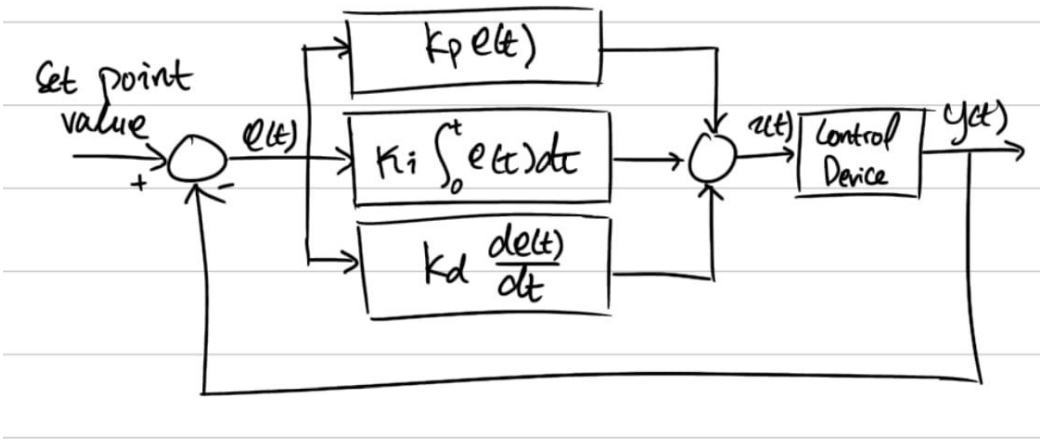


Figure 3. PID Loop (Src: Personal Documentation)

Lift can be optimized by positioning the hydrofoil at an angle relative to the water flow called the **angle of attack**. This angle **effects how much lift and drag (the force pulling to the back) created to the vessel**. The more lift, the faster it lifts the hull. To optimize lift to drag ratio, it depends on the shape of the foil. In a thin foil, the effective ratio which is 20 to 25:1, could be achieved with the angle of attack **3-4 degrees**. More than that, the lift increases slowly until 15 degrees where stalling become possible.

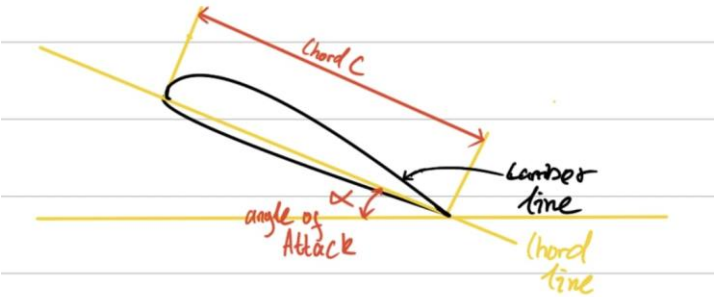


Figure 4. Angle of Attack (Src: Personal Documentation)

3. **The latest vessel from ITS MBST, Jalapatih VI, along with its specification. If you were tasked to designing a hydrofoil for this vessel, which foil profile would you use, and what would the configuration be (regarding quantity, shape, and placement of the vessel)? Please provide your reasoning.**

   Jalapatih VI is a solar boat with a top speed up to 30 km/h with 58 minutes of endurance. Other than that, it has a breadth/beam of 2.2m, Depth of 0.65m, drought of 0.17m, and an overall length (LOA) of 5.3m. The vessel, atleast from what I could see from its instagram post @its_msbt, consists of solar panels and its seat is slightly bit at front.

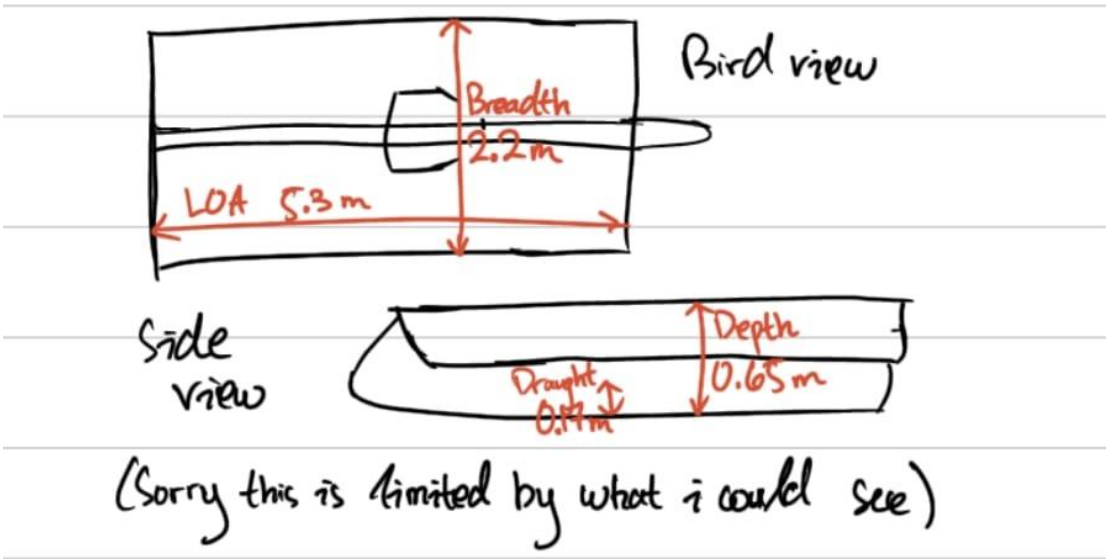


Figure 5. Vessel's Dimention (Src: Personal Documentation)

Considering the weight of the solar panels and the boatman, the **center of mass (COM) of the mass distribution of both the boatman and the solar panels are 2.62 meters**, assuming the approximate position of the seat (centered at 2,34m at the front), the surface density of the solar panels (10 kg/m^2), and the mass of the boatman (72 kg).

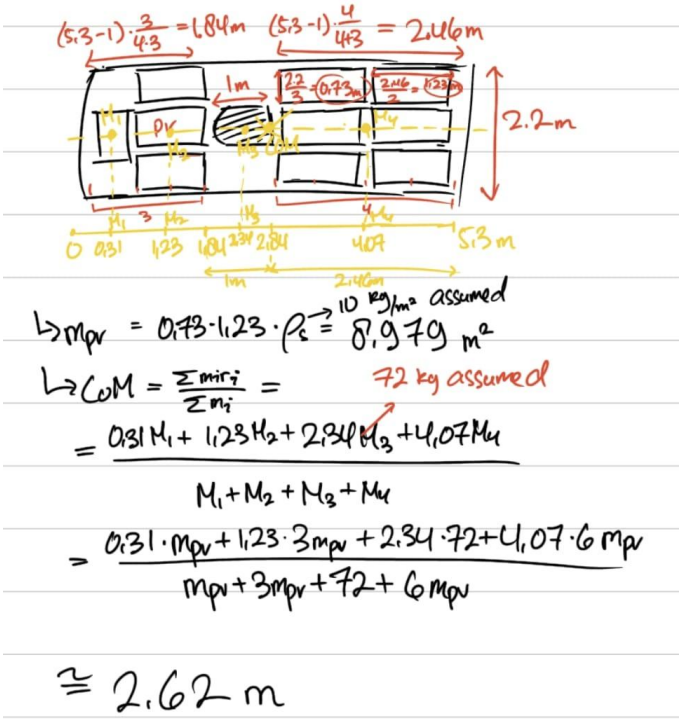


Figure 5. calculation of COM (Src: Personal Documentation)

Now lets make the struts connected to the hydrofoil 1 meter away from the front for the forward strut and 1 meter away from the back for the aft strut. The length between both struts (L) will consequently be 3.3 meters. The length between the forward strut and the COM (X) is 2.62 m minus 1 m, which is 1.62m. devide X by L, then we'll figure out the foil configuration for the vessel. Since the result is around 0.5, **a tandem configuration would fit our vessel to secure a balanced sailing**.

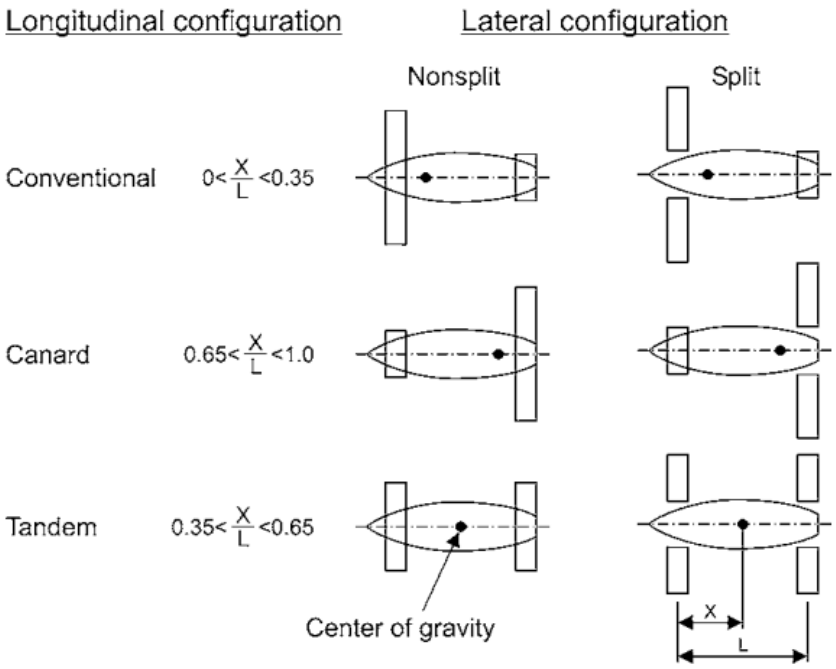


Figure 6. Foil Configuration (Src: Faltinsen, 2005)

There are also two types of categories of a hydrofoil in terms of how it submerged: free-surface piercing and fully submerged foil system. The free-surface piercing foil has its foil partially submerged and by diagonally orientate it. This hydrofoil could mechanically stabilize itself by its physical nature. But since it has larger displacement volume and surface, it creates bigger drag. On the other hand, the fully submerged foil only submerges its strut partially and its foils in a straight vertical orientation. This creates more difficulties on stabilizing it, but in return less drag. Since Jalapatih VI is a solar boat, which requires careful power management, **using the fully submerged foil would help to make the system efficient** wfith a good control system.

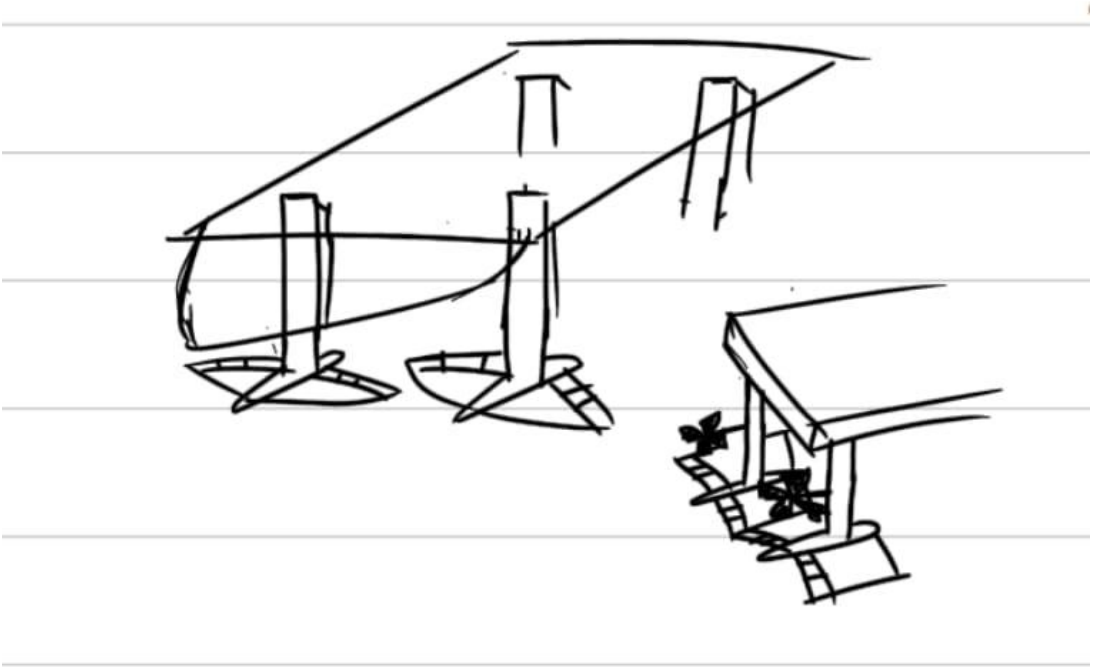


Figure 7. Hydrofoil's Rough Sketch

Sources:

1. Rosado, "How a Hydrofoil works". https://web.mit.edu/2.972/www/reports/hydrofoil/hydrofoil.html
2. Faltinsen, "Hydrodynamics of High-Speed Marine Vehicles"
3. Waterlust, "How do hydrofoil work – a deep dive into the physics" https://youtu.be/fzig8JR2iT0?si=nkgLrVzGP2tsEOxL